Quantum chemistry

Formula booklet

Chemistry S1-1 · From atomic entities to physico-chemical systems · handout chapters 1–5

How to read this booklet

Each row gives a quantity and where it is in the handout (§ section or table), the formula, then the symbols with their units and when the formula applies.

- † marks a result that is **not printed in the handout**: from the study guide or the worksheets, or a standard result for going further. Use it when it helps, but quote the handout's form when there is one.
- Numbers use the handout's values: $h = 6.62 \times 10^{-34}$ J s, $e = 1.6 \times 10^{-19}$ C, $c = 3.00 \times 10^{8}$ m s^{-1}, $R_H = 3.29 \times 10^{15}$ Hz.

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Constants

Speed of light	$c = 3.00 \times 10^{8}$ m s^{-1}
Planck constant	$h = 6.62 \times 10^{-34}$ J s
	$\hbar = h/2\pi = 1.05 \times 10^{-34}$ J s†
Elementary charge	$e = 1.6 \times 10^{-19}$ C
Electron mass	$m_e = 9.1094 \times 10^{-31}$ kg
Proton mass	$m_p = 1.6727 \times 10^{-27}$ kg
Neutron mass	$m_n = 1.6749 \times 10^{-27}$ kg
Atomic mass unit	1 amu $= 1.6605 \times 10^{-27}$ kg
Avogadro constant	$N_A = 6.022 \times 10^{23}$ mol^{-1}
Rydberg constant	$R_H = 3.29 \times 10^{15}$ Hz
	$R_H/c = 1.097 \times 10^{7}$ m^{-1}†
Bohr radius	$a_0 = 52.9$ pm
Coulomb constant	$k = 1/(4\pi\varepsilon_0) = 8.99 \times 10^{9}$ N m^2 C^{-2}
Vacuum permittivity	$\varepsilon_0 = 8.85 \times 10^{-12}$ F m^{-1}†
Ground state of H	$E_1 = -13.6$ eV

Conversions

1 eV $= 1.6 \times 10^{-19}$ J; 1 MeV $= 1.6 \times 10^{-13}$ J
1 eV/atom $= 96.5$ kJ mol^{-1}
E (eV) $\approx 1240/\lambda$ (nm) ($hc = 1240$ eV nm)
1 pm $= 10^{-12}$ m; 1 Å $= 100$ pm $= 10^{-10}$ m
1 fm $= 10^{-15}$ m $= 10^{-3}$ pm; 1 nm $= 10^{-9}$ m
1 D (debye) $= 3.336 \times 10^{-30}$ C m
1 amu $\times c^2 = 931.5$ MeV†

1 The atom and its particles (handout ch. 1)

Nuclide notation §1.3	${}^{A}_{Z}\mathrm{X}, \quad A = Z + N$	Z atomic number (protons), $N = A - Z$ neutrons, A mass number (nucleons). The neutral atom has Z electrons.
Ion §1.3	$q = (Z - n_e)\,e$	n_e number of electrons: X^{k+} has $Z - k$, X^{k-} has $Z + k$.
Isotopes; isoelectronic species §1.3	isotopes: same Z, different A	Isoelectronic: same number of electrons (F^-, Ne, Na^+, Mg^{2+}: 10).
Atomic mass unit §1.2.3	1 amu $= \frac{1}{12}\,m(^{12}\mathrm{C}) = 1.6605 \times 10^{-27}$ kg	$m_p = 1.0073$, $m_n = 1.0087$, $m_e = 1/1823$ amu (Table 1-1).
Mass of an atom §5.1	$m \approx A$ amu	Sum of protons and neutrons; the electrons are negligible.
Mole and molar mass†	$n = \dfrac{N}{N_A} = \dfrac{m}{M}$	N number of entities, m mass, M molar mass. Since $N_A \times 1$ amu $= 1$ g, M in g mol^{-1} equals the atomic mass in amu.
Specific charge (Thomson)† worksheet 1	$\dfrac{e}{m_e} = 1.76 \times 10^{11}$ C kg^{-1}	Charge-to-mass ratio from the deflection of cathode rays; e/m_p is 1836 times smaller.
Charge quantisation (Millikan) §1.2.3	$q = k\,e, \quad k \in \mathbb{Z}$	Every charge is a whole multiple of $e = 1.6 \times 10^{-19}$ C.

Coulomb's law†	$F = k\,\dfrac{\lvert q_1 q_2\rvert}{r^2}, \qquad E_p = k\,\dfrac{q_1 q_2}{r}$	$k = 8.99 \times 10^9$ N m^2 C^{-2}, r in m. $E_p > 0$: repulsion.
Closest approach (Rutherford)† worksheet 1	$d = k\,\dfrac{(2e)(Ze)}{E_k}$	Head-on α particle (charge $2e$, kinetic energy E_k) stops where E_k has become potential energy: an upper bound for the nucleus radius.
Density of a nucleus†	$\rho = \dfrac{m}{\frac{4}{3}\pi r^3}$	Nuclear matter: about 2×10^{17} kg m^{-3}.
Mass defect, binding energy† worksheet 1	$\Delta m = Z(m_p + m_e) + N m_n - m_{\text{atom}}$ $E = \Delta m\, c^2$	Δm in kg for E in J; 1 amu $\times\, c^2 = 931.5$ MeV.
Sizes §1.2.3	$r_{\text{atom}} \sim 100$ pm $r_{\text{nucleus}} \sim 10^{-3}$ to 10^{-2} pm	The nucleus is 10 000 to 100 000 times smaller than the atom.

2 Light and atomic spectra (handout ch. 2)

Electromagnetic radiation and photons

Wavelength, period, frequency §2.1	$\lambda = c\,T, \qquad \nu = \dfrac{1}{T}, \qquad c = \lambda\,\nu$	$c = 3.00 \times 10^8$ m s^{-1} in vacuum; λ in m, T in s, ν in Hz (s^{-1}).
Wavenumber†	$\tilde{\nu} = \dfrac{1}{\lambda} = \dfrac{\nu}{c}$	In m^{-1} (spectroscopy often uses cm^{-1}).
Photon energy (Planck) §2.2	$E = h\,\nu = \dfrac{h\,c}{\lambda}$	$h = 6.62 \times 10^{-34}$ J s. Shortcut: E (eV) $\approx 1240/\lambda$ (nm).
Photons in a beam† worksheet 2	$\dfrac{N}{t} = \dfrac{P}{h\,\nu} = \dfrac{P\,\lambda}{h\,c}$	P power in W, N/t photons per second. One mole of photons: $N_A\,h\nu$.

The hydrogen spectrum

Balmer series (visible) §2.4.3	$\nu = R_H\left(\dfrac{1}{2^2} - \dfrac{1}{n^2}\right), \quad n = 3, 4, 5, \ldots$	Lines at 656, 486, 434, 410 nm. The handout prints "$n = 1, 2, 3\ldots$": it must be $n \geq 3$.
Rydberg formula §2.4.3	$\nu = R_H\left(\dfrac{1}{n_1^2} - \dfrac{1}{n_2^2}\right), \quad n_1 < n_2$	$R_H = 3.29 \times 10^{15}$ Hz. As a wavelength: $\frac{1}{\lambda} = \frac{R_H}{c}\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$.
Series limit†	$\nu_{\lim} = \dfrac{R_H}{n_1^2}, \qquad \lambda_{\lim} = \dfrac{c\,n_1^2}{R_H}$	From $n_2 = \infty$ (ionised state) to n_1: the shortest wavelength of the series.
Number of lines†	$\dfrac{n\,(n-1)}{2}$	Lines that atoms excited up to level n can emit (every pair of levels).

Series	n_1	Region	First line ($n_2 = n_1 + 1$)	Limit ($n_2 = \infty$)
Lyman	1	UV	122 nm	91.2 nm
Balmer	2	visible	657 nm	365 nm
Paschen	3	IR	1876 nm	821 nm
Brackett†	4	IR	4.05 µm	1.46 µm
Pfund†	5	IR	7.46 µm	2.28 µm

Computed with $R_H = 3.29 \times 10^{15}$ Hz and $c = 3.00 \times 10^8$ m s^{-1} (measured Hα: 656 nm).

Bohr model and energy levels

Energy levels of H §2.5	$E_n = -\dfrac{13.6}{n^2}$ eV, $\quad n \in \mathbb{N}^*$	$n = 1$: ground state (−13.6 eV); $n > 1$: excited states; $n \to \infty$: $E = 0$, ionised. 13.6 eV $= h\,R_H$.
Hydrogen-like ion §3.3.1	$E_n = -\dfrac{13.6\,Z^2}{n^2}$ eV	One nucleus of charge Ze and one electron (He^+, Li^{2+}). Its lines: $\nu = Z^2 R_H\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$.
Transition §2.5	$\lvert\Delta E\rvert = \lvert E_{n_2} - E_{n_1}\rvert = h\,\nu = \dfrac{h\,c}{\lambda}$	Emission: the electron drops and a photon is emitted. Absorption: only a photon of exactly this energy lifts it.

Ionisation energy from level n†	$E_\mathrm{I} = \dfrac{13.6\,Z^2}{n^2}$ eV	H in its ground state: 13.6 eV = 1312 kJ mol^{-1}. If $h\nu > E_\mathrm{I}$ the electron leaves with $E_k = h\nu - E_\mathrm{I}$.
Size of an orbit or AO §3.3.3	$r = \dfrac{n^2}{Z}\,a_0, \qquad a_0 = 52.9$ pm	a_0: most probable distance between the nucleus and the 1s electron of H.

3 Quantum numbers and atomic orbitals (handout ch. 3)

The four quantum numbers

Principal n §3.2.1	$n = 1, 2, 3, \ldots$	Shell. Sets the energy (in H) and the size of the orbital.
Secondary ℓ §3.2.2	$0 \le \ell \le n-1$	Subshell: $\ell = 0, 1, 2, 3 \rightarrow$ s, p, d, f.
Magnetic m_ℓ §3.2.3	$-\ell \le m_\ell \le +\ell$	Orientation: $2\ell+1$ orbitals per subshell, of equal energy without a magnetic field.
Spin m_s §3.4.3	$m_s = +\frac{1}{2}\ (\uparrow) \quad$ or $\quad -\frac{1}{2}\ (\downarrow)$	Two electrons per orbital, with opposite spins.
Counting Table 4-1	orbitals: $2\ell+1$ per subshell, n^2 per shell electrons: $2(2\ell+1)$ per subshell, $2n^2$ per shell	Subshells hold 2 (s), 6 (p), 10 (d), 14 (f) electrons; shells 2, 8, 18, 32.

Orbitals

Atomic orbital (AO) §3.3	Ψ_{n,ℓ,m_ℓ} probability density $\lvert\Psi\rvert^2$	Probability of finding the electron in a small volume $\mathrm{d}V$: $\lvert\Psi\rvert^2\,\mathrm{d}V$. Boundary surface: 95 % of the probability.
Normalisation†	$\iiint_\mathrm{space} \lvert\Psi\rvert^2\,\mathrm{d}V = 1$	The electron is certainly somewhere.
Radial probability density† worksheet 3	$P(r) = 4\pi r^2\,\lvert\Psi(r)\rvert^2$	For an s orbital. Probability per unit distance of finding the electron at r; $\int_0^\infty P(r)\,\mathrm{d}r = 1$.
1s of a hydrogen-like species† worksheet 3	$P(r) = \dfrac{4Z^3}{a_0^3}\,r^2\,e^{-2Zr/a_0}$	From $\Psi_{1s} = \frac{1}{\sqrt{\pi}}\left(\frac{Z}{a_0}\right)^{3/2} e^{-Zr/a_0}$. Maximum at $r = a_0/Z$ (a_0 for H).
Nodes†	radial: $n-\ell-1$, angular: ℓ, total: $n-1$	Radial nodes are spheres, angular nodes planes (or cones); Ψ changes sign across a node (§3.3.3).
p orbitals Table 3-3	p_x, p_y, p_z: nodal planes (yOz), (xOz), (xOy)	$p_z \leftrightarrow m_\ell = 0$; p_x and p_y are combinations of $m_\ell = \pm 1$.

Many-electron atoms

Orbital approximation §3.4.1	$\Psi(1, 2, \ldots, N) = \Psi(1)\,\Psi(2)\cdots\Psi(N)$	Electron–electron repulsion neglected: each electron occupies its own AO.
Energy of an AO §3.4.2	$E = E(n, \ell, Z), \qquad E_{n\mathrm{s}} < E_{n\mathrm{p}} < E_{n\mathrm{d}}$	The levels of one shell are no longer degenerate.
Madelung's rule §3.4.2	energy increases with $n+\ell$ equal $n+\ell$: lower n first	Gives the filling order below.

Filling order: 1s < 2s < 2p < 3s < 3p < 4s < 3d < 4p < 5s < 4d < 5p < 6s < 4f < 5d < 6p < 7s < 5f < 6d < 7p

Sodium D doublet (spin) §3.4.3	$\Delta E = h\,c\left(\dfrac{1}{\lambda_1} - \dfrac{1}{\lambda_2}\right) \approx 2.1$ meV	$\lambda_1 = 589.0$ nm, $\lambda_2 = 589.6$ nm: the spin splits the 3p level in two.
Zeeman effect §3.2.3	a level ℓ splits into $2\ell+1$ sublevels	In a magnetic field B. Shift† $\Delta E = m_\ell\,\mu_B\,B$, with $\mu_B = 9.27 \times 10^{-24}$ J T^{-1}.

4 Electronic configuration and periodic table (handout ch. 4)

Aufbau principle §4.1.1	fill the AOs by increasing energy	In Madelung order. Gives the ground-state configuration.
Pauli exclusion principle §4.1.2	no two electrons share (n, ℓ, m_ℓ, m_s)	At most 2 electrons per AO, with opposite spins.

Subshell capacity Table 4-1	<table><tr><td></td><td>s</td><td>p</td><td>d</td><td>f</td></tr><tr><td>AOs</td><td>1</td><td>3</td><td>5</td><td>7</td></tr><tr><td>electrons</td><td>2</td><td>6</td><td>10</td><td>14</td></tr></table>	$2\ell+1$ AOs and $2(2\ell+1)$ electrons.
Hund's rule §4.1.3	maximum number of parallel spins in a subshell	Put one electron in each AO of the subshell (same m_s) before pairing. Unpaired electrons make the atom paramagnetic.
Configuration §4.1.5	X: [previous noble gas] + valence electrons	Cl: $[\mathrm{Ne}]\,3s^2\,3p^5$; Fe: $[\mathrm{Ar}]\,3d^6\,4s^2$. Core electrons: the noble-gas part.
Ions†	cations lose ns before $(n-1)$d	Fe^{2+}: $[\mathrm{Ar}]\,3d^6$; Fe^{3+}: $[\mathrm{Ar}]\,3d^5$. Anions fill the next free AOs.
Exceptions (d block) §4.2.2	Cr: $[\mathrm{Ar}]\,3d^5\,4s^1$ Cu: $[\mathrm{Ar}]\,3d^{10}\,4s^1$	A half-filled or full d subshell is more stable (also Mo, Ag, Au).
Reading the table §4.2	period = n of the valence shell block = subshell being filled	s block: groups 1–2 and He; p block: 13–18; d block: 3–12; f block: lanthanides and actinides.
Valence electrons and group §4.2.2	s, p blocks: N_{val} = position in the period d block: $(n-1)\mathrm{d}^x\,n\mathrm{s}^2$, x = group − 2	Groups 13–18 have 3–8 valence electrons. In the d block, valence electrons = group number (Fe: group 8). He: $1s^2$ but group 18.
Period lengths†	2, 8, 8, 18, 18, 32, 32	Twice the number of AOs filled in the period: s; s+p; s+d+p; s+f+d+p.

5 Atomic properties and periodicity (handout ch. 5)

Atomic mass

Average atomic mass §5.1.2	$\bar{M} = \sum_i x_i\,M_i, \qquad \sum_i x_i = 1$	x_i natural abundance (fraction) of isotope i, M_i its mass in amu. Cl: $0.7577 \times 34.97 + 0.2423 \times 36.97 = 35.45$ amu.
Two isotopes: abundance† worksheet 5	$x_1 = \dfrac{M_2 - \bar{M}}{M_2 - M_1}$	From $\bar{M} = x_1 M_1 + (1-x_1)\,M_2$.

Effective nuclear charge and radius

Effective nuclear charge† study guide	$Z_{\mathrm{eff}} = Z - \sigma$	σ: screening constant, the shielding by the other electrons.
Slater's rules (ns or np electron)† study guide	$\sigma = 0.35\,N_n + 0.85\,N_{n-1} + 1.00\,N_{\le n-2}$	N_n: other electrons of shell n (0.30 in 1s); N_{n-1}: electrons of shell $n-1$; $N_{\le n-2}$: deeper ones. Cl: $\sigma = 10.9$, $Z_{\mathrm{eff}} = 6.1$; Na: $Z_{\mathrm{eff}} = 2.2$.
Size trend†	$r \propto \dfrac{n^2}{Z_{\mathrm{eff}}}$	From $r = n^2 a_0/Z$ with $Z \to Z_{\mathrm{eff}}$: smaller across a period, larger down a group.
Measured radii §5.2.1	$r_{\mathrm{cov}} = \frac{1}{2}\,d_{\mathrm{A-A}}, \qquad r_{\mathrm{met}} = \frac{1}{2}\,d_{\mathrm{metal}}$ $r_{\mathrm{vdW}} = \frac{1}{2}\,d_{\text{non-bonded}}$	Ionic crystal: $d = r_+ + r_-$. $r_{\mathrm{cation}} < r_{\mathrm{atom}} < r_{\mathrm{anion}}$; $r_{\mathrm{vdW}} > r_{\mathrm{cov}}$.

Ionisation energy

First ionisation energy §5.3	$\mathrm{X(g)} \longrightarrow \mathrm{X^+(g)} + \mathrm{e^-}$	$E_{\mathrm{I}} > 0$: minimum energy to remove the outermost electron, in $\mathrm{kJ\,mol^{-1}}$. Successive energies: $E_{\mathrm{I}1} < E_{\mathrm{I}2} < E_{\mathrm{I}3} \ldots$
eV per atom to kJ per mole†	$E\,(\mathrm{kJ\,mol^{-1}}) = \dfrac{E\,(\mathrm{J}) \times N_A}{1000}$	1 eV/atom = 96.5 $\mathrm{kJ\,mol^{-1}}$. H: 13.6 eV = 1312 $\mathrm{kJ\,mol^{-1}}$.
Photoelectron spectroscopy §5.3.1	$E_{\mathrm{I}} = h\,\nu - E_k, \qquad E_k = \frac{1}{2}\,m_e v^2$	A photon of known energy $h\nu$ ejects an electron; its kinetic energy E_k gives the binding energy of its AO.

Polarity, polarisability, electronegativity

Dipole moment §5.4	$\vec{\mu} = q\,\vec{d}$	Two charges $\pm q$ a distance d apart; arrow from $-q$ to $+q$. Unit: C m or D ($1\ \mathrm{D} = 3.336 \times 10^{-30}$ C m).
Partial charge (ionic character)†	$\delta = \dfrac{\mu}{e\,d}$	d bond length. δ in units of e: 0 for a covalent bond, 1 for a fully ionic one (ionic character = $100\,\delta$ %).
Induced dipole†	$\vec{\mu}_{\mathrm{ind}} = \alpha\,\vec{E}$	α polarisability: large for big, diffuse electron clouds.

Pauling electronegativity† §5.5.1, Table 5-3	$\lvert\chi_A - \chi_B\rvert = 0.102\sqrt{\Delta}$ $\Delta = D_{AB} - \sqrt{D_{AA}\,D_{BB}}$	D bond dissociation energies in kJ mol^{-1}; reference $\chi_{\mathrm{H}} = 2.20$. H–F: $\Delta = 304.5$, $\lvert\Delta\chi\rvert = 1.78$.
Allred–Rochow electronegativity† §5.5.1	$\chi = 0.359\,\dfrac{Z_{\mathrm{eff}}}{r^2} + 0.744$	r covalent radius in Å. Cl: $Z_{\mathrm{eff}} = 6.1$, $r = 0.99$ Å, $\chi = 2.98$.
Bond polarity §5.5	$\Delta\chi = 0$: non-polar; $\Delta\chi > 0$: polar $\Delta\chi \gtrsim 2$: ionic	The more electronegative atom carries $\delta-$.

Trends in the periodic table

Property	Across a period (→)	Down a group (↓)	Reason
Z_{eff} (valence)	increases	almost constant	poor screening inside a shell
Atomic radius	decreases	increases	$r \propto n^2/Z_{\mathrm{eff}}$
Ionisation energy	increases (dips: Be > B, N > O)	decreases	electron held more strongly by a larger Z_{eff} at a smaller r
Electronegativity	increases (F: 3.98)	decreases	same as ionisation energy
Polarisability	decreases	increases	follows the radius

Anomalies: Be > B because B loses a 2p electron, higher in energy than 2s; N > O because O loses a paired 2p electron (pairing repulsion).
Isoelectronic series: the radius decreases as Z increases ($O^{2-} > F^- > Ne > Na^+ > Mg^{2+}$).

6 Going further: standard quantum results†

None of these is printed in the handout. They are the usual companions of the quantum model and appear in many exercises.

de Broglie wavelength†	$\lambda = \dfrac{h}{p} = \dfrac{h}{m\,v}$	Any particle is also a wave. Electron accelerated by a voltage U: $\lambda = h/\sqrt{2m_e eU}$.
Photon momentum†	$p = \dfrac{h}{\lambda} = \dfrac{E}{c}$	A photon has no mass but carries momentum.
Heisenberg uncertainty†	$\Delta x\,\Delta p_x \geq \dfrac{\hbar}{2}$	Position and momentum cannot both be known exactly: no classical orbit.
Photoelectric effect†	$E_{k,\max} = h\,\nu - W_0, \qquad \nu_0 = \dfrac{W_0}{h}$	W_0 work function of the metal; no electron below the threshold ν_0.
Bohr: quantised angular momentum†	$m_e\,v\,r = n\,\hbar$	With Coulomb's force as the centripetal force, it gives the three results below.
Bohr radius and speed†	$a_0 = \dfrac{\varepsilon_0 h^2}{\pi m_e e^2}$ $v_n = \dfrac{Z}{n}\dfrac{e^2}{2\varepsilon_0 h} = \dfrac{Z}{n} \times 2.19 \times 10^6\ \mathrm{m\,s^{-1}}$	$a_0 = 52.9$ pm, $r_n = n^2 a_0/Z$.
Bohr energy and R_H†	$E_n = -\dfrac{m_e e^4}{8\varepsilon_0^2 h^2}\dfrac{Z^2}{n^2}, \qquad R_H = \dfrac{m_e e^4}{8\varepsilon_0^2 h^3}$	$m_e e^4/(8\varepsilon_0^2 h^2) = 13.6$ eV and $R_H = 3.29 \times 10^{15}$ Hz.
Schrödinger equation†	$-\dfrac{\hbar^2}{2m}\nabla^2\Psi + V\,\Psi = E\,\Psi$	Its solutions for H are the AOs Ψ_{n,ℓ,m_ℓ} and the energies E_n.
Particle in a box (1D)†	$E_n = \dfrac{n^2 h^2}{8mL^2}, \qquad \Psi_n(x) = \sqrt{\dfrac{2}{L}}\,\sin\dfrac{n\pi x}{L}$	Box of length L, $n = 1, 2, 3\ldots$: confinement quantises the energy.
Mulliken electronegativity†	$\chi_M = \dfrac{E_{\mathrm{I}} + E_{\mathrm{ea}}}{2}$	E_{ea} electron affinity; both in eV.

Summary

Units of all quantities

Wavelength	λ	m (nm for light)
Frequency	ν	hertz, Hz $= \mathrm{s}^{-1}$
Wavenumber	$\tilde{\nu}$	m^{-1} (cm^{-1})
Energy of a photon or level	E	J, or eV per atom
Ionisation energy	E_I	$\mathrm{kJ\,mol}^{-1}$ (chemistry)
Rydberg constant	R_H	Hz (as in the handout)
Planck constant	h	J s

Radius, distance	$r,\ a_0$	m (pm or Å)
Atomic mass	m	amu (or kg)
Molar mass	M	$\mathrm{g\,mol}^{-1}$
Dipole moment	μ	C m, or debye (D)
Bond energy	D_{AB}	$\mathrm{kJ\,mol}^{-1}$
Electronegativity	χ	no unit
Quantum numbers	$n,\ \ell,\ m_\ell,\ m_s$	no unit

Common slips

ν (frequency, Greek nu) is not v (speed). R_H is in Hz here; in m^{-1} it is R_H/c. Energies per atom (eV, J) and per mole ($\mathrm{kJ\,mol}^{-1}$) differ by N_A. Wavelengths go in metres in $E = hc/\lambda$.